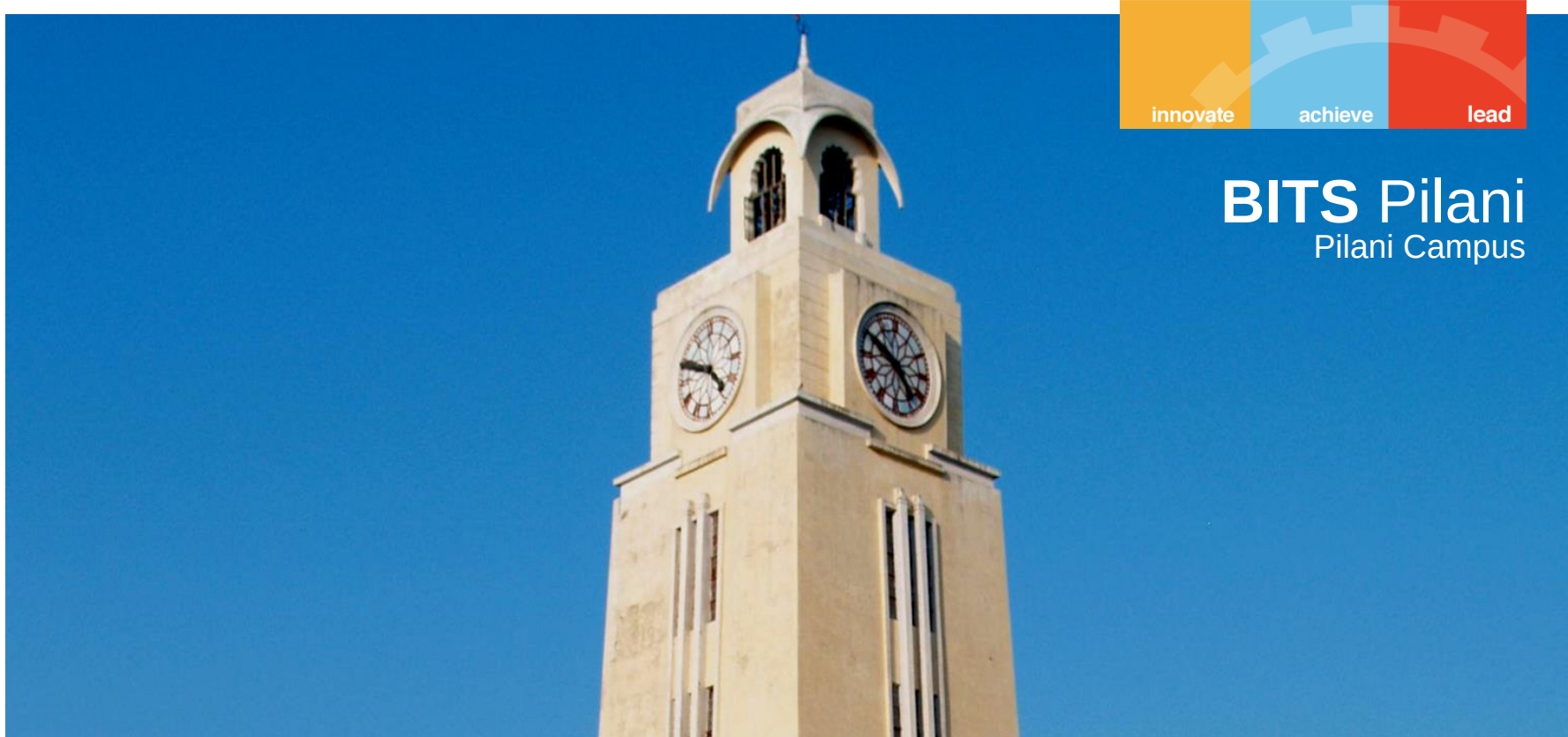




CHEM F111 : General Chemistry

Semester II: AY 2023-24

Lecture- 15, 19th February 2024, Monday

Last Class:

- Franck Condon principles; shapes of bands and intensities;
- Transition moments; selection rules for molecular transitions;
- Beer Lamber law; mixture of two absorbing species;
- Schematics of uv-vis and fluorescence spectrometers;
- Vibrational spectroscopy –introduction;
- Vibration frequencies, vibrational energy of a diatomic molecule;
- Gross selection rule; specific selection rule considering SHO model;
- Anharmonicity and overtones.

Recap: Vibrational spectroscopy



$$V = \frac{1}{2} k x^2 = \frac{1}{2} k (R - R_e)^2$$

In the harmonic approximation:

$$E_v = \left(v + \frac{1}{2} \right) h\nu; \quad v = 0, 1, 2, \dots$$

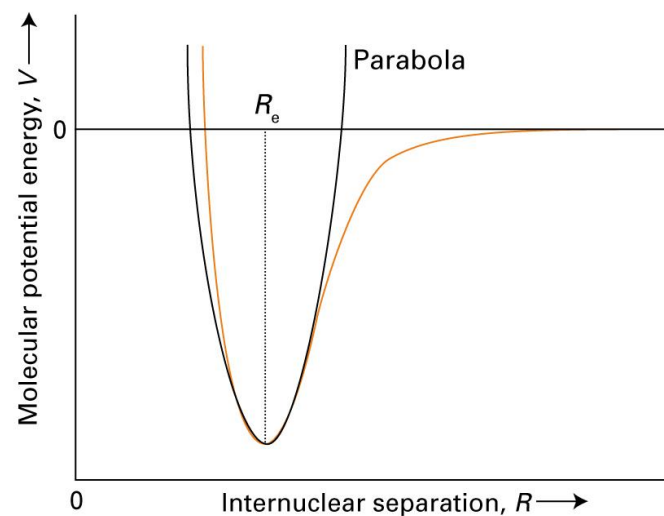
$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}; \quad \mu = \frac{m_A m_B}{m_A + m_B}$$

Anharmonicity:

$$E_v = (v + \frac{1}{2})hc\bar{\nu} - (v + \frac{1}{2})^2 hc\bar{\nu} x_e + \dots$$

x_e is anharmonicity constant.

Additional weak absorption lines or **overtones corresponding to $\Delta v = 2, 3, \dots$ appear.** (near IR)



.....Continued

Vibrational spectroscopy
(IR spectroscopy)



For a molecule with N atoms, the number of coordinates required to specify the position of all the atoms is $3N$.

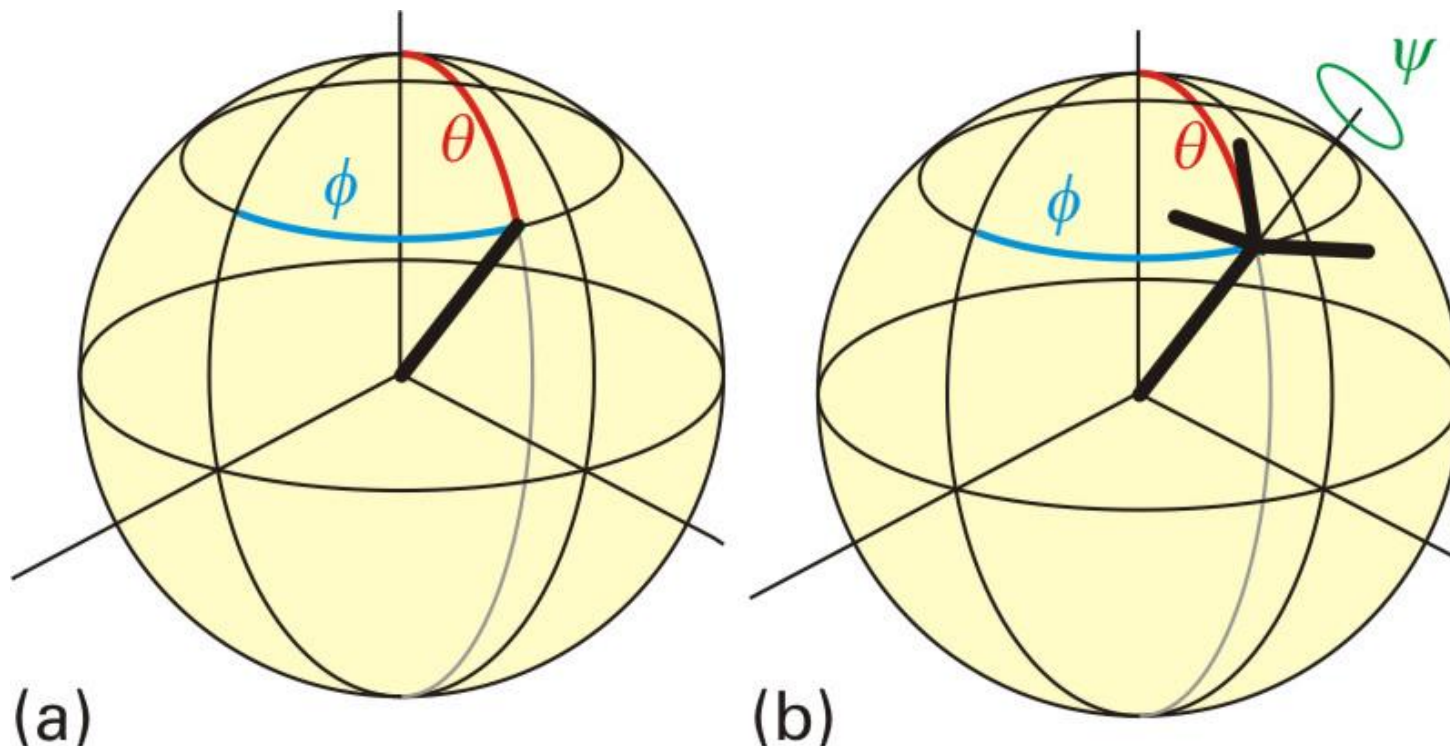
These $3N$ displacements can be thought of in terms of various degrees of freedom.

Translational : 3

Rotational : 2 for linear and 3 for nonlinear molecule.

Vibrational : $3N - 5$ for linear and $3N - 6$ for nonlinear molecule.

Degrees of freedom



The orientation of a linear molecule requires the specification of two angles (the latitude and longitude of its axis).

The orientation of a non-linear molecule requires the specification of three angles (the latitude and longitude of its axis and the angle of twist (the torsion angle) around that axis).

Normal Modes:

Description of vibration is most convenient and simple in terms of normal modes.

Normal mode – Collective, independent, synchronous motion of group of atoms, that may be excited without exciting any other mode.

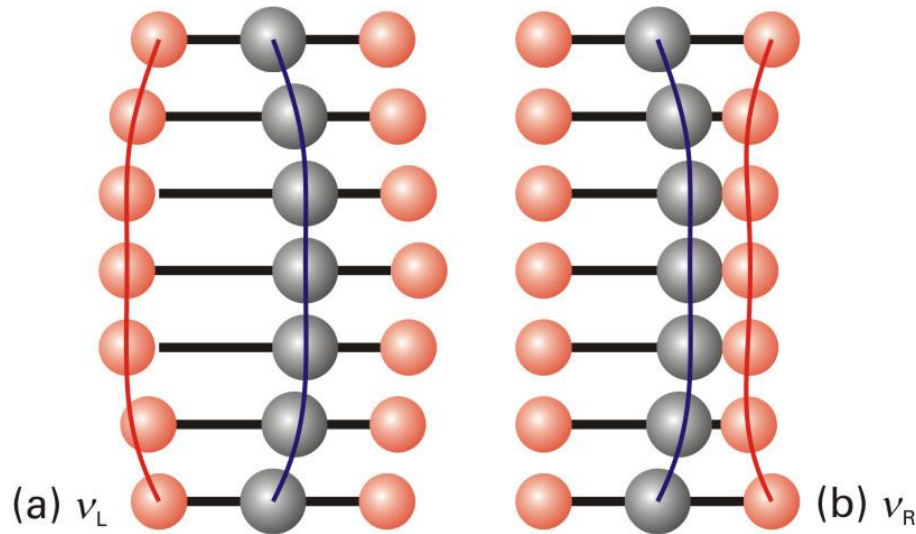
Each normal mode may be approximated as a harmonic oscillator with a characteristic frequency and an effective mass.

Any arbitrary displacement may be expressed as a superposition of the normal modes.

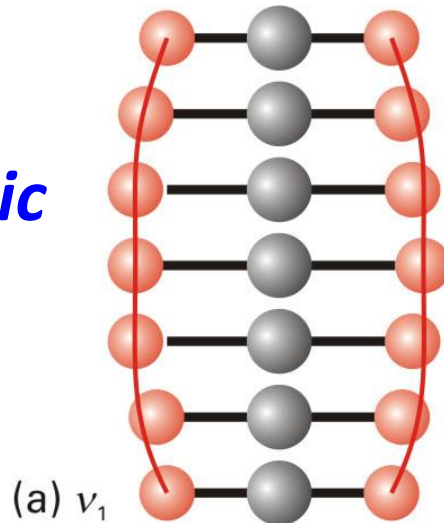
Superposition of normal modes



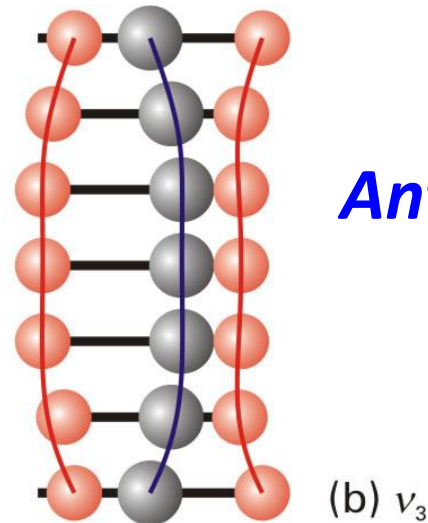
The stretching vibrations of a CO_2 molecule



Symmetric stretch



Antisymmetric stretch



CO₂: Normal mode of vibrations



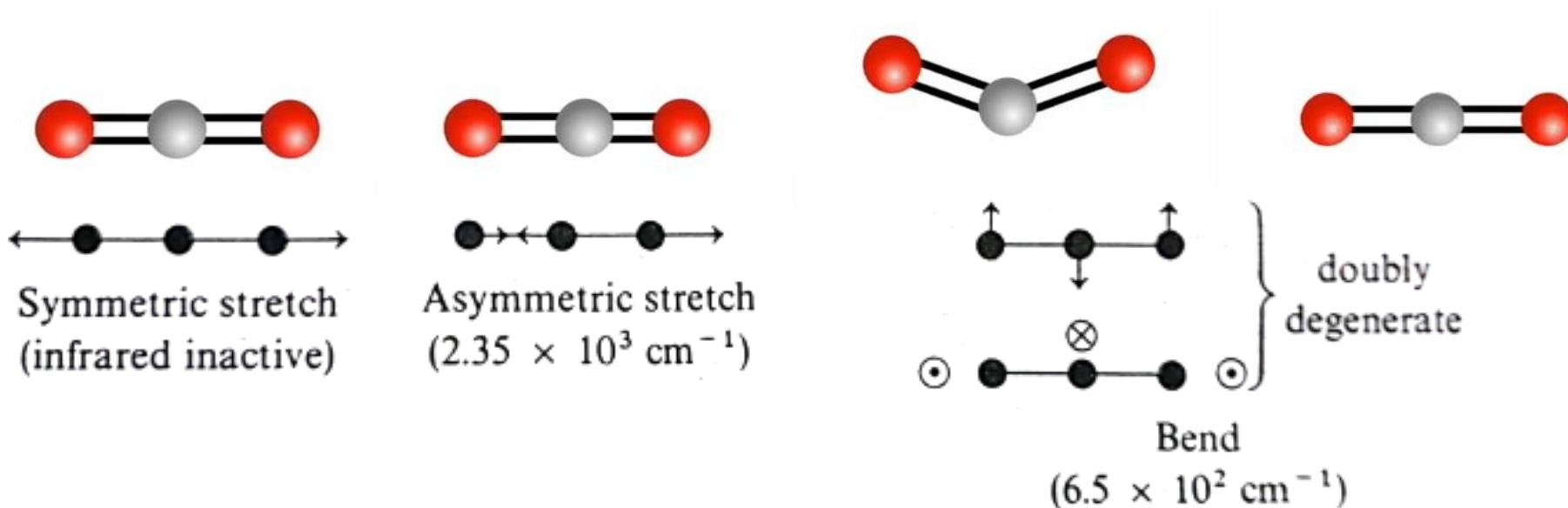
For CO₂ Normal modes are $3N-5 = (3 \times 3) - 5 = 4$;

*2 stretching; (Stretching modes are equal to no of bonded connections; remaining are bending modes);
In CO₂ these are also called as parallel modes (dipole moment change is parallel to internuclear axis);*

In stretching bond length changes;

In bending bond angle changes (low freq. compared to stretching) and for CO₂ there are two bending modes (perpendicular modes; degenerate; combined).

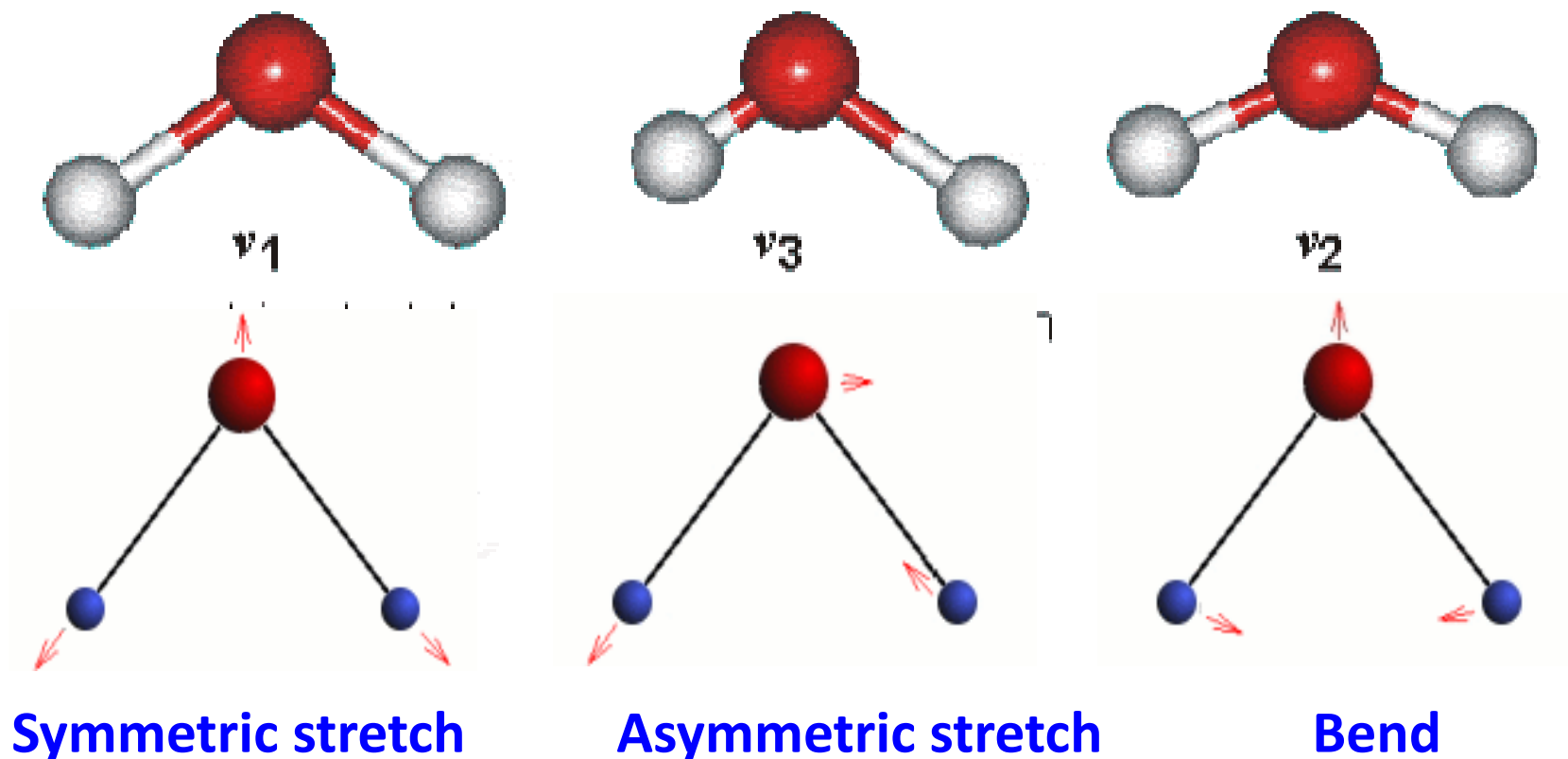
CO₂: Normal mode of vibrations



H₂O: Normal modes of vibrations

For H₂O Normal modes are $3N-6 = (3 \times 3) - 6 = 3$;

2 stretching and one bending;



Vibrational transition of a molecule:

Gross selection rule



The Gross Selection Rule: The vibrational motion corresponding to normal mode must give rise to a changing dipole moment (for IR active)

Collective vibrations of whole molecule occur at around 500-1500 cm^{-1} . Fingerprint region.

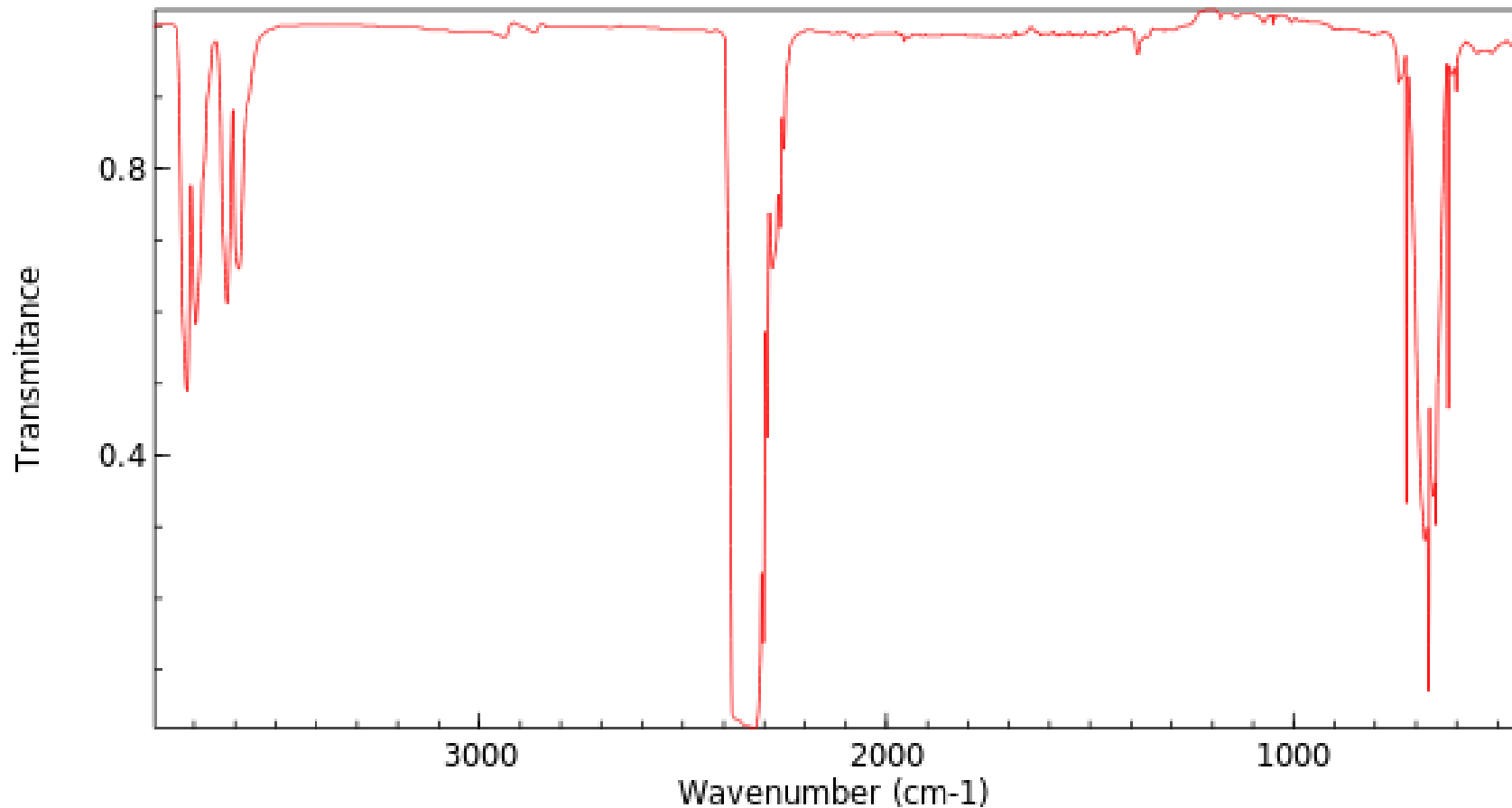
Example: Bending modes of CO_2 are also located in the Fingerprint region. Difficult to identify.

Green House effect; Global warming.

The vibrational spectra of CO₂ molecule



CARBON DIOXIDE
INFRARED SPECTRUM

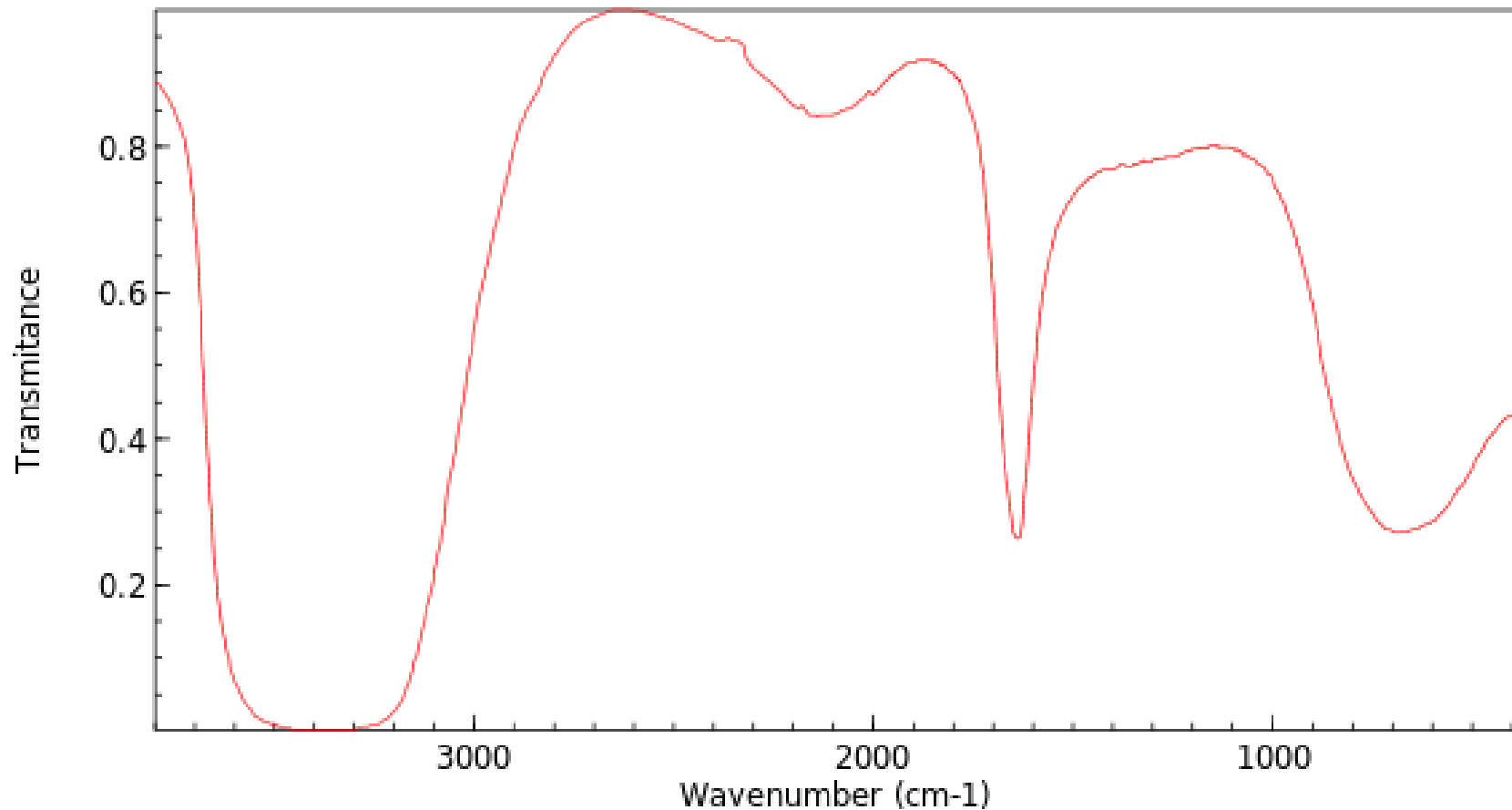


NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)

The vibrational spectra of H₂O molecule



WATER
INFRARED SPECTRUM



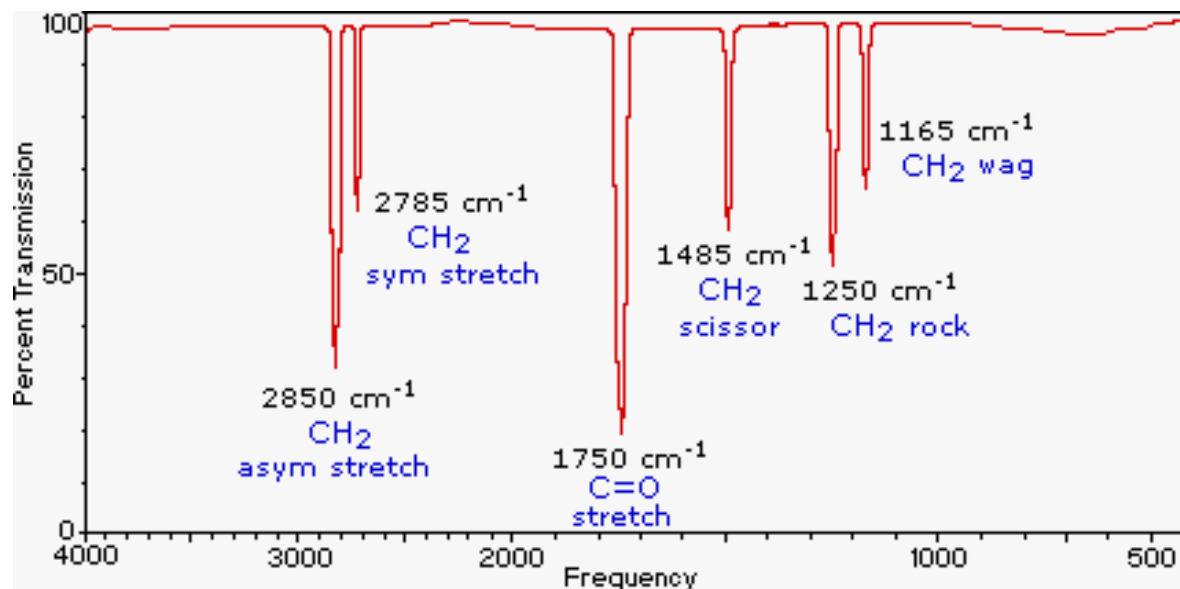
NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)

Vibrational spectroscopy (polyatomic)



- Infrared spectroscopy is a powerful tool in identifying organic molecules.*
- Some modes essentially describe the motions of the individual functional groups.*

Gas Phase Infrared Spectrum of Formaldehyde, $\text{H}_2\text{C}=\text{O}$



IR frequencies of functional groups

GROUP	BOND	FREQUENCY RANGE (CM ⁻¹)
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Alkyl	C—H	2853–2962
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Alcohol	O—H	3590–3650
---------	-----	-----------

Amine	N—H	3300–3500
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$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}; \quad \mu = \frac{m_A m_B}{m_A + m_B}$$

BOND	FREQUENCY RANGE (CM ⁻¹)
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C≡C	2100–2260
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C≡N	2220–2260
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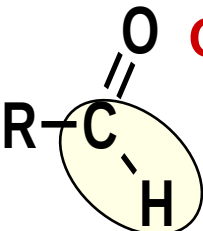
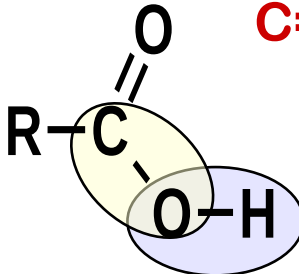
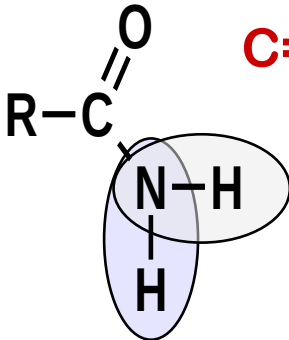
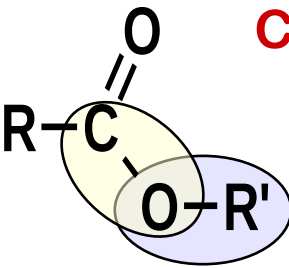
C=C	1620–1680
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C=O	1630–1780
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Here the effect is
mainly due to bond
strength

Interpreting IR spectra



 C=O at 1725 cm⁻¹ also look for aldehyde CH 2850 and 2750 cm⁻¹	 C=O at 1710 cm⁻¹ also look for OH (H-bonded) and C-O ~1200 cm⁻¹
 C=O at 1690 cm⁻¹ also look for two NH peaks at 3400 cm⁻¹	 C=O at 1735 cm⁻¹ also look for two C-O at 1200 and 1000 cm⁻¹

Ketones have **C=O at 1715 cm⁻¹** and **no NH, OH, C-O or -CHO**

Anhydrides have **two C=O peaks near 1800 cm⁻¹** and **two C-O**

IR frequency data

innovate

achieve

lead

Group	Frequency Range (cm ⁻¹)	Intensity ^a	
A. Alkyl			
C—H (stretching)	2853–2962	(m–s)	
Isopropyl, —CH(CH ₃) ₂	1380–1385	(s)	
	and 1365–1370	(s)	
<i>tert</i> -Butyl, —C(CH ₃) ₃	1385–1395	(m)	
	and ~ 1365	(s)	
B. Alkenyl			
C—H (stretching)	3010–3095	(m)	
C=C (stretching)	1620–1680	(v)	
R—CH=CH ₂	985–1000	(s)	
R ₂ C=CH ₂	(out-of-plane C—H bendings) and	905–920	(s)
		880–900	(s)
<i>cis</i> -RCH=CHR	675–730	(s)	
<i>trans</i> -RCH=CHR	960–975	(s)	
C. Alkynyl			
≡C—H (stretching)	~ 3300	(s)	
C≡C (stretching)	2100–2260	(v)	
D. Aromatic			
Ar—H (stretching)	~ 3030	(v)	
Aromatic substitution type (C—H out-of-plane bendings)			
Monosubstituted	690–710	(very s)	
<i>o</i> -Disubstituted	and 730–770	(very s)	
<i>m</i> -Disubstituted	735–770	(s)	
	680–725	(s)	
	and 750–810	(very s)	
<i>p</i> -Disubstituted	800–860	(very s)	

E. Alcohols, Phenols, and Carboxylic Acids

O—H (stretching)

Alcohols, phenols (dilute solutions)	3590–3650	(sharp, v)
Alcohols, phenols (hydrogen bonded)	3200–3550	(broad, s)
Carboxylic acids (hydrogen bonded)	2500–3000	(broad, v)

F. Aldehydes, Ketones, Esters, and Carboxylic

C=O (stretching)	1630–1780	(s)
Aldehydes	1690–1740	(s)
Ketones	1680–1750	(s)
Esters	1735–1750	(s)
Carboxylic acids	1710–1780	(s)
Amides	1630–1690	(s)

G. Amines

N—H	3300–3500	(m)
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H. Nitriles

C≡N	2220–2260	(m)
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^aAbbreviations: s = strong, m = medium, w = weak, v = variable, ~ = approximately.

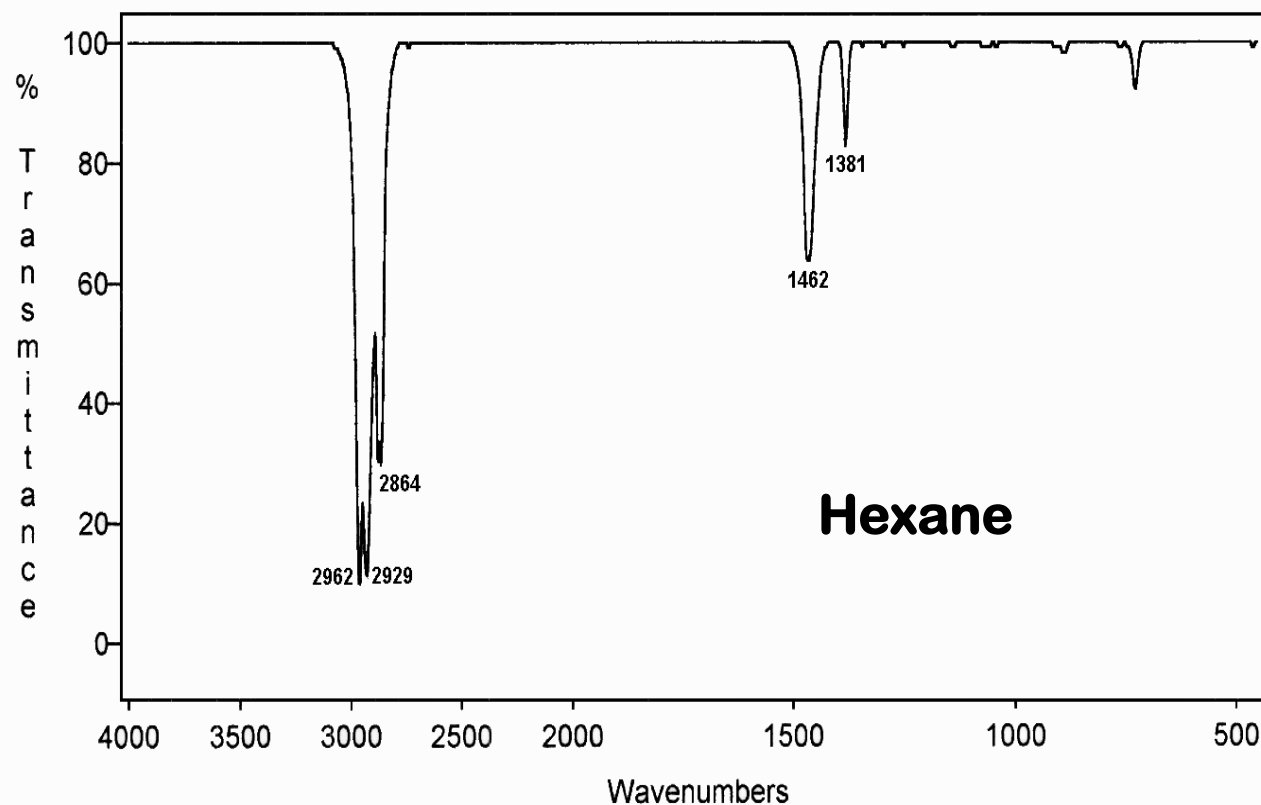
Interpreting IR spectra



C-H stretching frequencies:

<3000 cm^{-1} for sp^3 ; >3000 cm^{-1} for sp^2 ; $\sim 3300 \text{ cm}^{-1}$ for sp ;

Methylene and methyl C-H stretching: $\sim 2850 \text{ cm}^{-1}$ (sym); $\sim 2950 \text{ cm}^{-1}$ (asym)

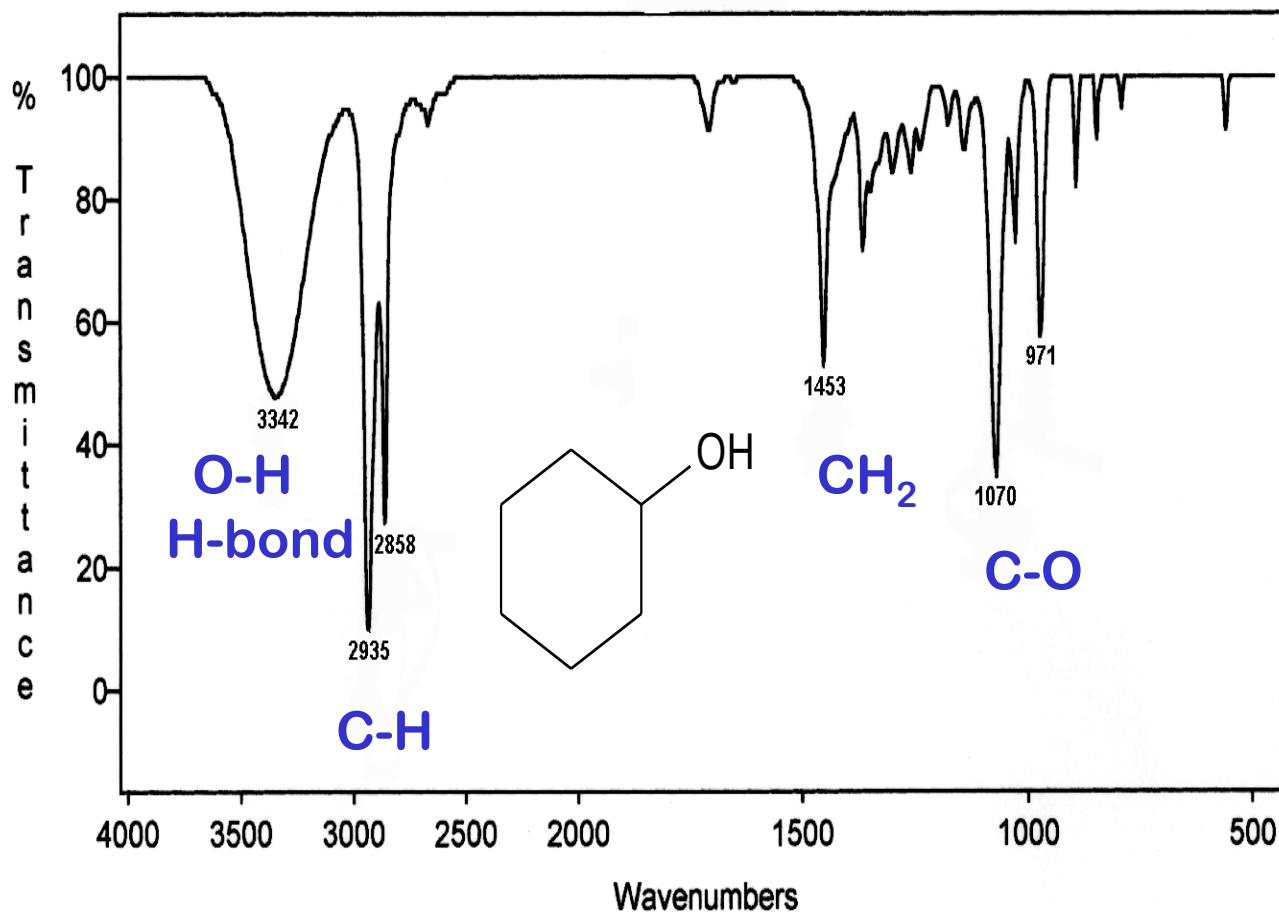


Interpreting IR spectra

O-H stretching: $\sim 3600\text{ cm}^{-1}$ for alcohols;

broadens and lowers to 3300 cm^{-1} due to H-bonding in alcohols, acids, etc.

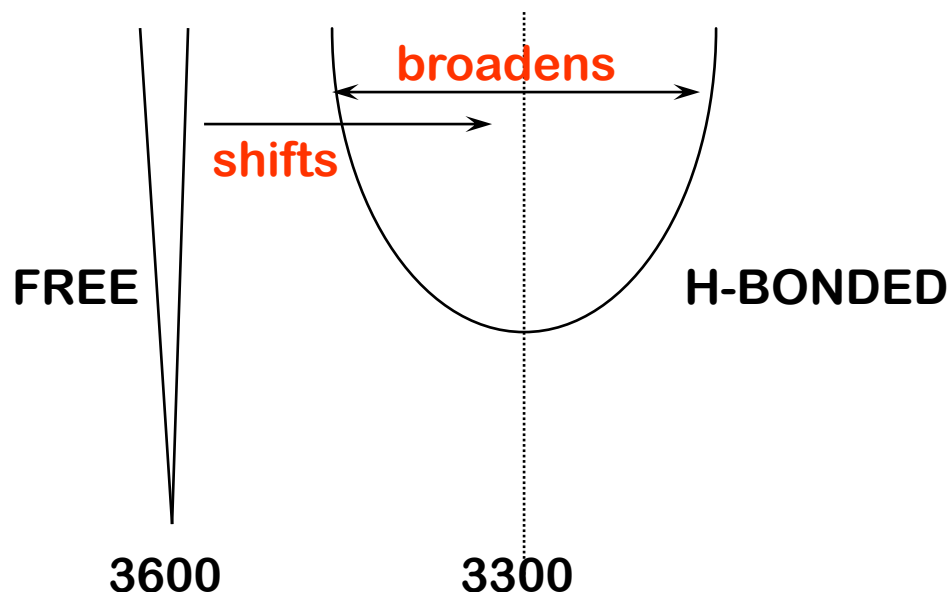
How to distinguish between intramolecular and intermolecular H-bonding?



The O-H stretching region



- O-H 3600 cm^{-1} (alcohol, free)
- O-H 3300 cm^{-1} (alcohols & acids, H-bonding)



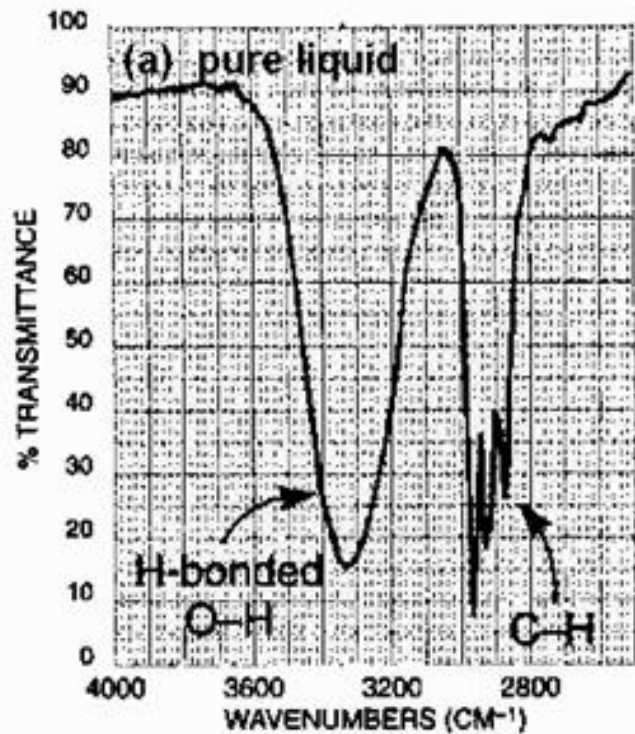
- **O-H stretching frequency decreases due to hydrogen bonding.**

Effect of Dilution on IR of 1-Butanol

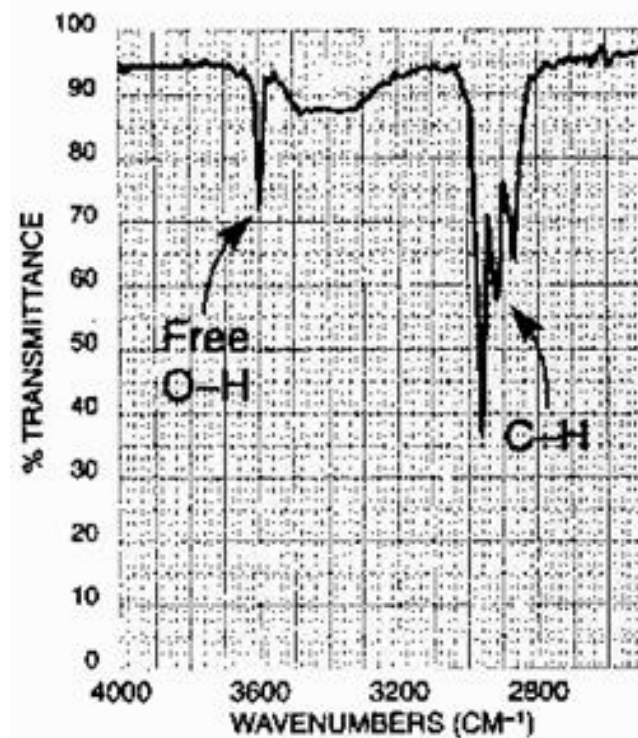
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achieve

lead



Pure Liquid



Very Dilute Solution

1-Butanol

Dilution in a non-Hydrogen bonding solvent